

BRAIN GATE: Assistive Technology For Productive Living



Dr Rajiv Kumar Singh holds a PhD in electronics engineering from IIT (BHU), Varanasi, and is currently employed with Institute of Engineering & Technology, Lucknow as assistant professor. He has received senior research fellowship of UGC at Centre of Advanced Study at IIT, Varanasi. His interests include wired and wireless technologies for high-speed Internet access

Vivek Kumar Singh is pursuing BTech in electronics and instrumentation engineering from Institute of Engineering & Technology, Lucknow. His current research interests include biomedical instrumentation

Brain gate is a new technological development in the field of medical electronics that enables people with severe paralysis or neurological problems to live more productively and independently

Brain gate is a brain implanted technology that was first introduced in 2003. It was developed by a bio-tech company called Cyberkinetics, in association with Department of Neuroscience at Brown University, USA. It is a brain-computer interface (BCI) for people who have lost control of their limbs due to any accidental cause, such as spinal cord injury or Amyotrophic Lateral Sclerosis (ALS). It is an interfacing of brain with a computer for productive and independent life of paralysed people, to help them control a computer using their thoughts.

In case of damaged organs or paralysis, the brain is functional, generating signals and sending these to the respective terminals (organs). But, due to paralysis or a damaged organ, control signals are not able to reach their destination(s). A computer chip, known as pedestal, is implanted into the brain. It senses intentions of the user and generates commands accordingly.

In a brain gate system, intentions of the user are converted into electrical signals. These are further decoded using a program that generates control signals to perform tasks like moving a robotic arm or a wheel chair.

The biological neural system

A human brain consists of a complex structure of 100 billion neurons connected to each other through synaptic terminals (Fig. 1). Thoughts processed by the brain are governed by the neurological network of neurons. Dendrites sense neural activities and collectively transfer it to soma, which is a part of the neuron cell body that sums up all dendrites signals. If it is above a particular threshold—which is a minimum signal required to process the action—it is transferred

to synopsis through axons (which includes a number of myelin sheath). Synopsis of one neuron is interconnected with dendrites of the other neuron, which, through this signal, is transferred to other neurons, and this process continues.

There are two types of BCI.

One-way BCI. In this mode, information is transferred in only one direction, that is, from brain to computer.

Two-way BCI. In this mode, information exchange occurs in both directions between the brain and the computer.

Basic elements of brain gate

A brain gate implantation system contains a pedestal, connector, neuron signal interpreter and monitor. A pedestal contains a silicon chip, which has 100 hair-thin microelectrodes. Typical dimensions of a pedestal chip are 4mm × 4mm × 1mm. It is fixed in the motor cortex region of the human brain.

A patient cable connector wire connects a pedestal and an interpreter to convey neuron signals produced by the brain. This signal is further amplified. A neuron signal interpreter converts it into an electrical signal, which can be used to control actions; for example, moving a robotic arm, wheel chair, or the cursor on a computer monitor.

Executing operations using BCI

If a person thinks of doing something, then her or his neural impulse varies

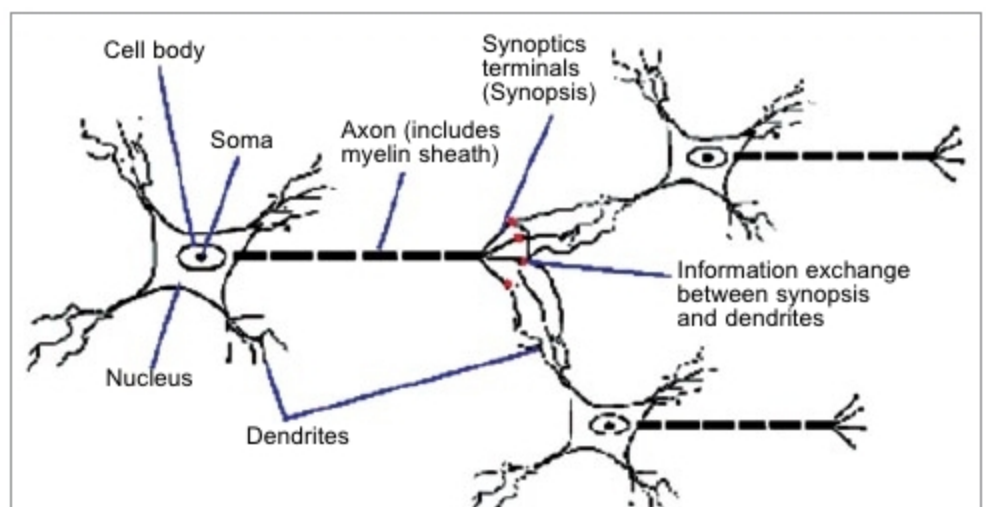


Fig. 1: Biological neural network